

3.03 Forces and Newton's Laws

3.03a	Newton's first law	a) Understand the concept and vector nature of a force. <i>A force has both a magnitude and direction and can cause an object with a given mass to change its velocity.</i> <i>Includes using directed line segments to represent forces (acting in at most two dimensions).</i> <i>Learners should be able to identify the forces acting on a system and represent them in a force diagram.</i>		MR1
3.03b		b) Understand and be able to use Newton's first law. <i>A particle that is at rest (or moving with constant velocity) will remain at rest (or moving with constant velocity) until acted upon by an external force.</i> <i>Learners should be able to complete a diagram with the force(s) required for a given body to remain in equilibrium.</i>		

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
3.03c 3.03e 3.03d	Newton's second law	c) Understand and be able to use Newton's second law ($F = ma$) for motion in a straight line for bodies of constant mass moving under the action of constant forces. <i>e.g. A car moving along a road, a passenger riding in a lift or a crane lifting a weight.</i> <i>For stage 1 learners, examples can be restricted to problems in which the forces acting on the body will be collinear, in two perpendicular directions or given as 2-D vectors.</i> d) Understand and be able to use Newton's second law ($F = ma$) in simple cases of forces given as two dimensional vectors. <i>e.g. Find in vector form the force acting on a body of mass 2 kg when it is accelerating at $(4\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$.</i> <i>Questions set involving vectors may involve either column vector notation $\mathbf{F} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$ or $\mathbf{i}, \mathbf{j}$ notation</i> $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j}$.	e) Be able to extend use of Newton's second law to situations where forces need to be resolved (restricted to two dimensions). <i>e.g. A force acting downwards on a body at a given angle to the horizontal or the motion of a body projected down a line of greatest slope of an inclined plane.</i>	MR2

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3.03f	Weight	f) Understand and be able to use the weight ($W = mg$) of a body to model the motion in a straight line under gravity. <i>e.g. A ball falling through the air.</i>		MR3
3.03g		g) Understand the gravitational acceleration, g, and its value in S.I. units to varying degrees of accuracy. <i>The value of g may be assumed to take a constant value of 9.8 ms^{-2} but learners should be aware that g is not a universal constant but depends on location in the universe.</i> <i>[The inverse square law for gravitation is not required.]</i>		

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3.03h 3.03l 3.03i 3.03j 3.03k 3.03m	Newton's third law	h) Understand and be able to use Newton's third law. <i>Every action has an equal and opposite reaction.</i> <i>Learners should understand and be able to use the concept that a system in which none of its components have any relative motion may be modelled as a single particle.</i> i) Understand and be able to use the concept of a normal reaction force. <i>Learners should understand and use the result that when an object is resting on a horizontal surface the normal reaction force is equal and opposite to the weight of the object. This includes knowing that when $R = 0$ contact is lost.</i> j) Be able to use the model of a 'smooth' contact and understand the limitations of the model. k) Be able to use the concept of equilibrium together with one dimensional motion in a straight line to solve problems that involve connected particles and smooth pulleys. <i>e.g. A train engine pulling a train carriage(s) along a straight horizontal track or the vertical motion of two particles, connected by a light inextensible string passing over a fixed smooth peg or light pulley.</i>	l) Be able to extend use of Newton's third law to situations where forces need to be resolved (restricted to two dimensions). m) Be able to use the principle that a particle is in equilibrium if and only if the sum of the resolved parts in a given direction is zero. <i>Problems may involve the resolving of forces, including cases where it is sensible to:</i> <i>1. resolve horizontally and vertically,</i> <i>2. resolve parallel and perpendicular to an inclined plane,</i> <i>3. resolve in directions to be chosen by the learner, or</i> <i>4. use a polygon of forces.</i>	MR4

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3.03n 3.03o	Newton's third law (continued)	n) Be able to solve problems involving simple cases of equilibrium of forces on a particle in two dimensions using vectors, including connected particles and smooth pulleys. <i>e.g. Finding the required force $\mathbf{F}$ for a particle to remain in equilibrium when under the action of forces $\mathbf{F}_1, \mathbf{F}_2, \dots$</i> <i>For stage 1 learners, examples can be restricted to problems in which the forces acting on the body will be collinear, in two perpendicular directions or given as 2-D vectors.</i>	o) Be able to resolve forces for more advanced problems involving connected particles and smooth pulleys. <i>e.g. The motion of two particles, connected by a light inextensible string passing over a light pulley placed at the top of an inclined plane.</i>	MR4
3.03p	Applications of vectors in a plane		p) Understand the term 'resultant' as applied to two or more forces acting at a point and be able to use vector addition in solving problems involving resultants and components of forces. <i>Includes understanding that the velocity vector gives the direction of motion and the acceleration vector gives the direction of resultant force.</i> <i>Includes being able to find and use perpendicular components of a force, for example to find the resultant of a system of forces or to calculate the magnitude and direction of a force.</i> <i>[Solutions will involve calculation, not scale drawing.]</i>	MR5

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3.03q			q) Be able to solve problems involving the dynamics of motion for a particle moving in a plane under the action of a force or forces. <i>e.g. At time t s the force acting on a particle P of mass 4 kg is $(4\mathbf{i} + t\mathbf{j})$ N. P is initially at rest at the point with position vector $(3\mathbf{i} - 5\mathbf{j})$. Find the position vector of P when $t = 3$ s.</i>	

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3.03r 3.03s 3.03t 3.03u 3.03v	Frictional forces	r) Understand the concept of a frictional force and be able to apply it in contexts where the force is given in vector or component form, or the magnitude and direction of the force are given.	s) Be able to represent the contact force between two rough surfaces by two components (the 'normal' contact force and the 'frictional' contact force). <i>Questions set will explicitly use the terms normal (contact) force, frictional (contact) force and magnitude of the contact force.</i> t) Understand and be able to use the coefficient of friction and the $F \leq \mu R$ model of friction in one and two dimensions, including the concept of limiting friction. <i>[Knowledge of the angle of friction is excluded.]</i> u) Understand and be able to solve problems regarding the static equilibrium of a body on a rough surface and solve problems regarding limiting equilibrium. v) Understand and be able to solve problems regarding the motion of a body on a rough surface. <i>e.g. The motion of a body projected down a line of greatest slope on a rough inclined plane.</i> <i>[Problems set on inclined planes will only consider motion along the line of greatest slope and therefore a vector consideration of the motion will not be required.]</i>	MR6

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